ML101
Self-Organizing Maps
ITA Institute of Technology in Architecture

Faculty of Architecture / ETH Zurich
caad.arch.ethz.ch/blog

November 12, 2026

Self Organizing Maps

Feature Vector + Euclidean Distance

A Common Ground for Comparing Objects

(color, size, weight, sweetness)

(0 red, 12.3 cm, 180 g, 2 very)

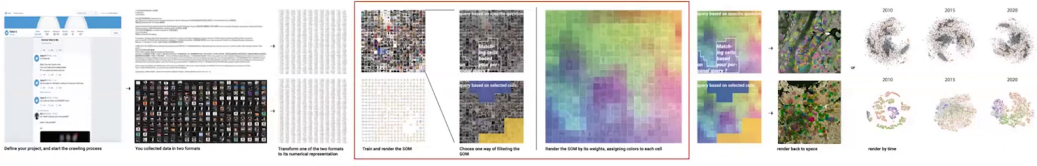
A

(color, size, weight, sweetness)

(1 orange, 11.7 cm, 170 g, 1 yes)

B

$$\|AB\| = \sqrt{(1-0)^2 + (11.7-12.3)^2 + (170-180)^2 + (2-1)^2} = 10.1173..$$


Define your project, and start the crawling process
You collected data in two formats
Transform one of the two formats to its numerical representation
Train and render the SOM
Choose one way of filtering the SOM
Render the SOM by its weights, assigning colors to each cell
render back to space
2010
2015
2020
render by time

FEATURE VECTORS AND EUCLIDEAN DISTANCE

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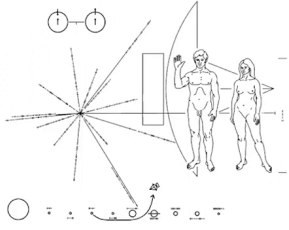
Feature Vector

Numerical features that represent some object

An apple

(color, size, weight, sweetness)

(0 red, 12.3 cm, 180 g, 2 very)



@Pioneer plaque

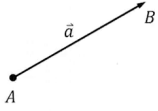
A person

(sex, age, weight, height)

(0 male, 25, 70 kg, 182 cm)

Euclidean distance

distance between two points in Euclidean space



$$\|AB\| = \sqrt{(B_1 - A_1)^2 + (B_2 - A_2)^2 + \cdots + (B_n - A_n)^2} = \|\vec{a}\|$$

Feature Vector + Euclidean Distance

A Common Ground for Comparing Objects

(color, size, weight, sweetness)

(0 red, 12.3 cm, 180 g, 2 very)

A

(color, size, weight, sweetness)

(1 orange, 11.7 cm, 170 g, 1 yes)

B

$$\|AB\| = \sqrt{(1-0)^2 + (11.7-12.3)^2 + (170-180)^2 + (2-1)^2} = 10.1173..$$

SELF-ORGANIZING MAP (SOM)

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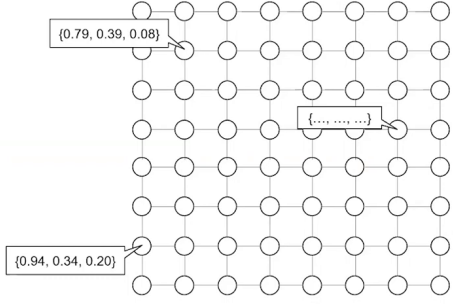
Self-organizing map

Kohonen, 1982



SOM is a grid of feature vectors (cells / units)
above is an example of 8 x 8 x 3 SOM
where 8 x 8 is the size, 3 is the dimensionality of feature vectors

WHAT DOES SOM DO

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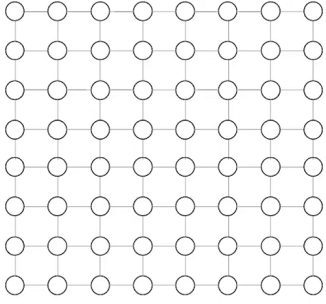
A Projection from Given Data on Regular Grid

A “compressed” representation of our data



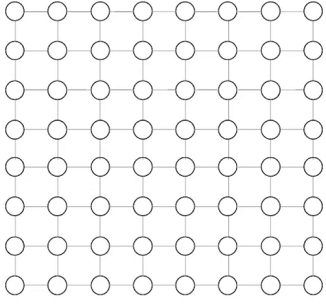
Each SOM cell represents a cluster of similar items from our dataset

A Projection from Given Data on Regular Grid

A "compressed" representation of our data

Our Dataset

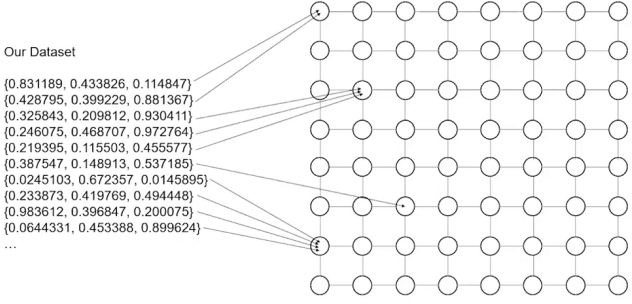
{0.831189, 0.433826, 0.114847}
{0.428795, 0.399229, 0.881367}
{0.325843, 0.209812, 0.930411}
{0.246075, 0.468707, 0.972764}
{0.219395, 0.115503, 0.455577}
{0.387547, 0.148913, 0.537185}
{0.0245103, 0.672357, 0.0145895}
{0.233873, 0.419769, 0.494448}
{0.983612, 0.396847, 0.200075}
{0.0644331, 0.453388, 0.899624}
…



Each SOM cell represents a cluster of similar items from our dataset

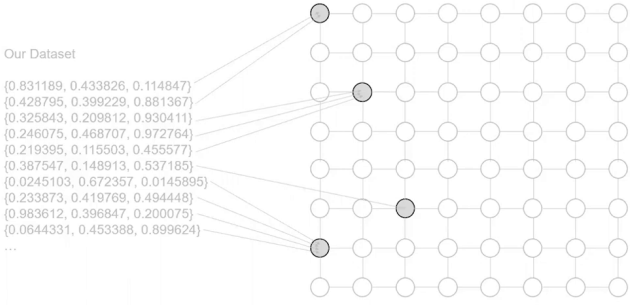
A Projection from Given Data on Regular Grid

A "compressed" representation of our data



Each SOM cell represents a cluster of similar items from our dataset

The Best Matching Unit (BMU)



For the items of our dataset, their corresponding SOM cells are called their BEST MATCHING UNIT

MORE IN DETAIL

Training the SOM

A Huge Dataset Impossible to Manage Manually

For example 100k feature vectors of apples

(color, size, weight, sweetness)

(0 red, 12.3 cm, 180 g, 2 very)

(1 orange, 11.7 cm, 170 g, 1 yes)

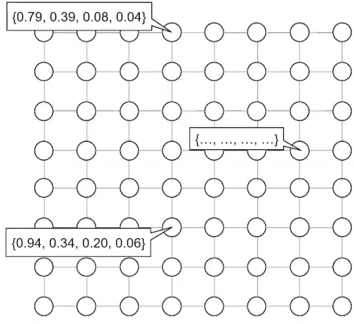
... (..., ..., ..., ...)

But we know some of them are similar to each other

A Random SOM

Our dataset

{0.837001,0.727544,0.0394684,0.299028}
{0.684913,0.337634,0.892134,0.794974}
{0.178787,0.837747,0.2533,0.182625}
{0.352355,0.611677,0.643471,0.486624}
{0.0623343,0.330299,0.440126,0.550363}
{0.95928,0.453402,0.250339,0.138137}
{0.906858,0.0194981,0.464387,0.836483}
{0.70246,0.532199,0.463251,0.0710004}
{0.84464,0.606172,0.864091,0.26712}
{0.427036,0.299789,0.128026,0.759577}
…

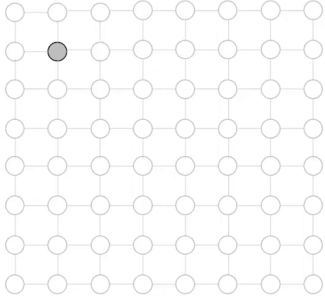


We create a SOM with its feature vectors randomly initialized

Find BMU

Our dataset

{0.837001,0.727544,0.0394684,0.299028}
{0.684913,0.337634,0.892134,0.794974}
{0.178787,0.837747,0.2533,0.182625}
{0.352355,0.611677,0.643471,0.486624}
{0.0623343,0.330299,0.440126,0.550363}
{0.95928,0.453402,0.250339,0.138137}
{0.906858,0.0194981,0.464387,0.836483}
{0.70246,0.532199,0.463251,0.0710004}
{0.84464,0.606172,0.864091,0.26712}
{0.427036,0.299789,0.128026,0.759577}
…



We randomly select one item from our dataset, find its BMU

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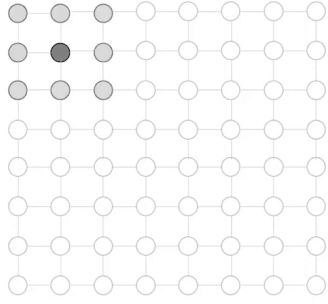
Update BMU Feature Vector

"*Stretching*" the SOM in feature space

Our dataset

{0.837001,0.727544,0.0394684,0.299028}
{0.684913,0.337634,0.892134,0.794974}
{0.178787,0.837747,0.2533,0.182625}
{0.352355,0.611677,0.643471,0.486624}
{0.0623343,0.330299,0.440126,0.550363}
{0.95928,0.453402,0.250339,0.138137}
{0.906858,0.0194981,0.464387,0.836483}
{0.70246,0.532199,0.463251,0.0710004}
{0.84464,0.606172,0.864091,0.26712}
{0.427036,0.299789,0.128026,0.759577}
…



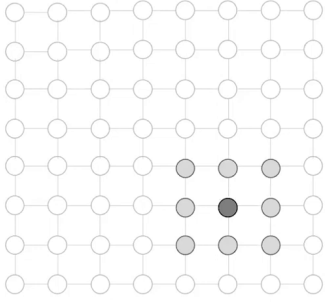
We update the BMU and its neighbors' feature vectors so that they are similar to our selected element
The update is stronger to the BMU than to its neighbors

Update BMU Feature Vector

"*Stretching*" the SOM in feature space

Our dataset

{0.837001,0.727544,0.0394684,0.299028}
{0.684913,0.337634,0.892134,0.794974}
{0.178787,0.837747,0.2533,0.182625}
{0.352355,0.611677,0.643471,0.486624}
{0.0623343,0.330299,0.440126,0.550363}
{0.95928,0.453402,0.250339,0.138137}
{0.906858,0.0194981,0.464387,0.836483}
{0.70246,0.532199,0.463251,0.0710004}
{0.84464,0.606172,0.864091,0.26712}
{0.427036,0.299789,0.128026,0.759577}
…



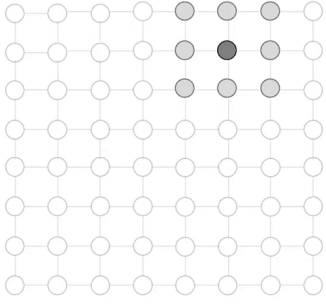
We move to another randomly selected item, and repeat the process

Update BMU Feature Vector

"*Stretching*" the SOM in feature space

Our dataset

{0.837001,0.727544,0.0394684,0.299028}
{0.684913,0.337634,0.892134,0.794974}
{0.178787,0.837747,0.2533,0.182625}
{0.352355,0.611677,0.643471,0.486624}
{0.0623343,0.330299,0.440126,0.550363}
{0.95928,0.453402,0.250339,0.138137}
{0.906858,0.0194981,0.464387,0.836483}
{0.70246,0.532199,0.463251,0.0710004}
{0.84464,0.606172,0.864091,0.26712}
{0.427036,0.299789,0.128026,0.759577}
…



Repeating the process over the entire dataset called an Epoch
We do many Epochs with the radius of neighbor cells decreasing

EXAMPLES OF SOMS ON DIFFERENT TYPES OF DATA

We can render SOM cells by the items they contain

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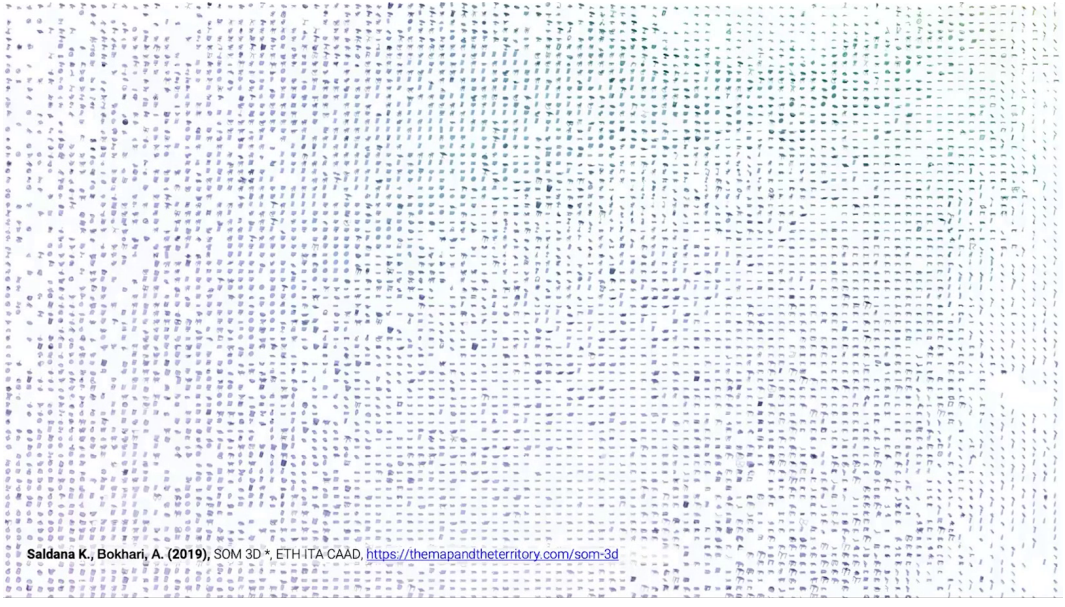
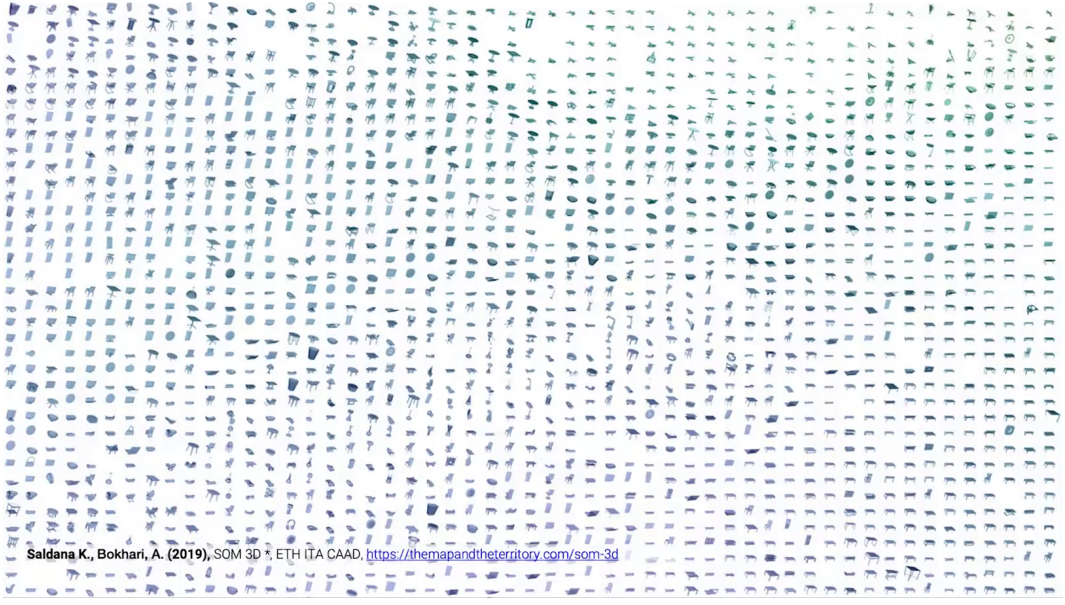
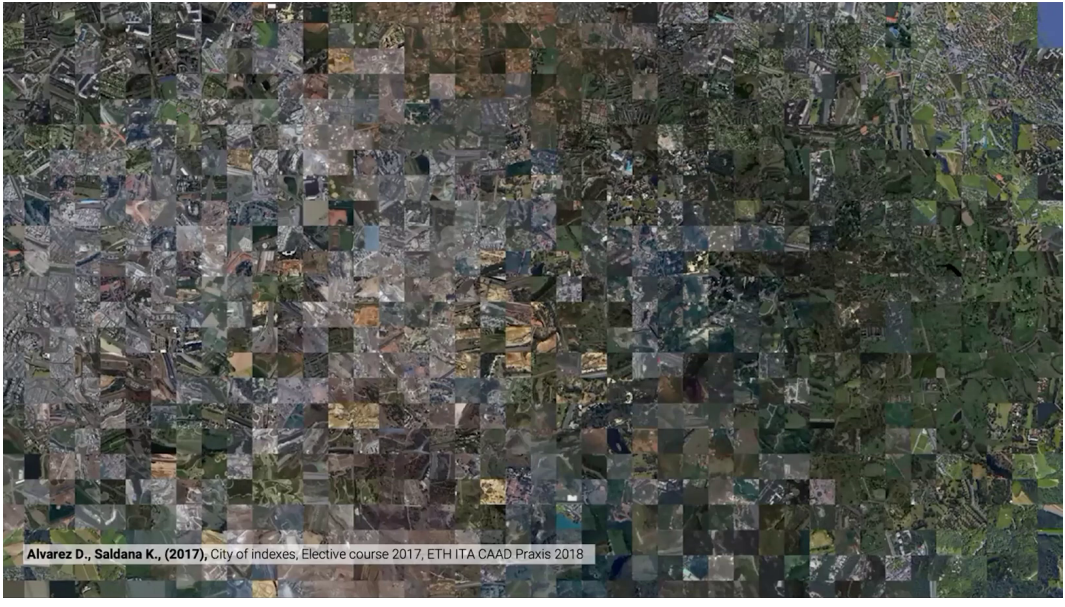
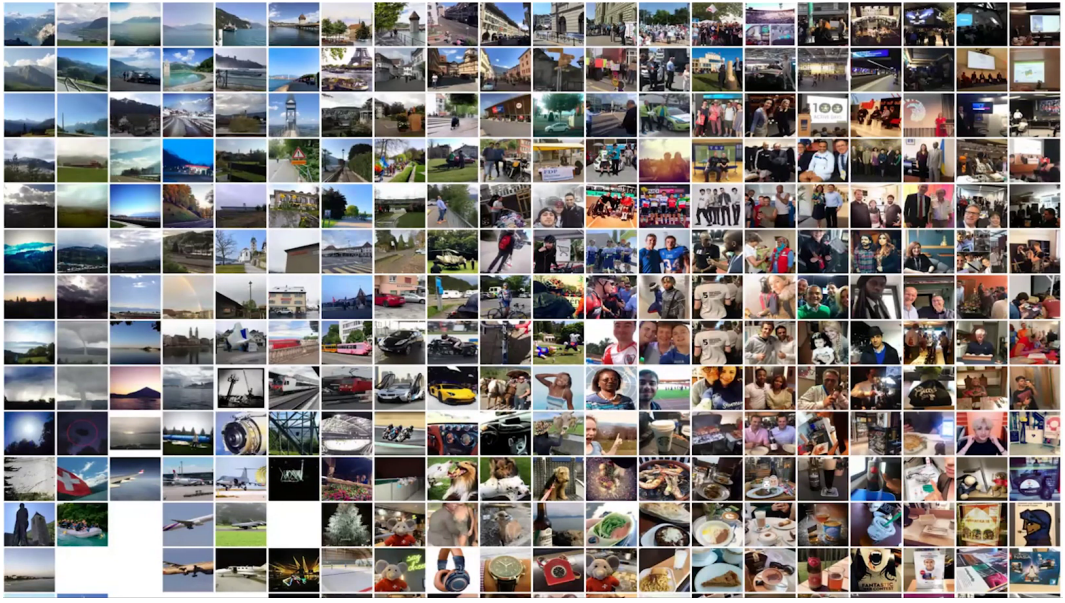
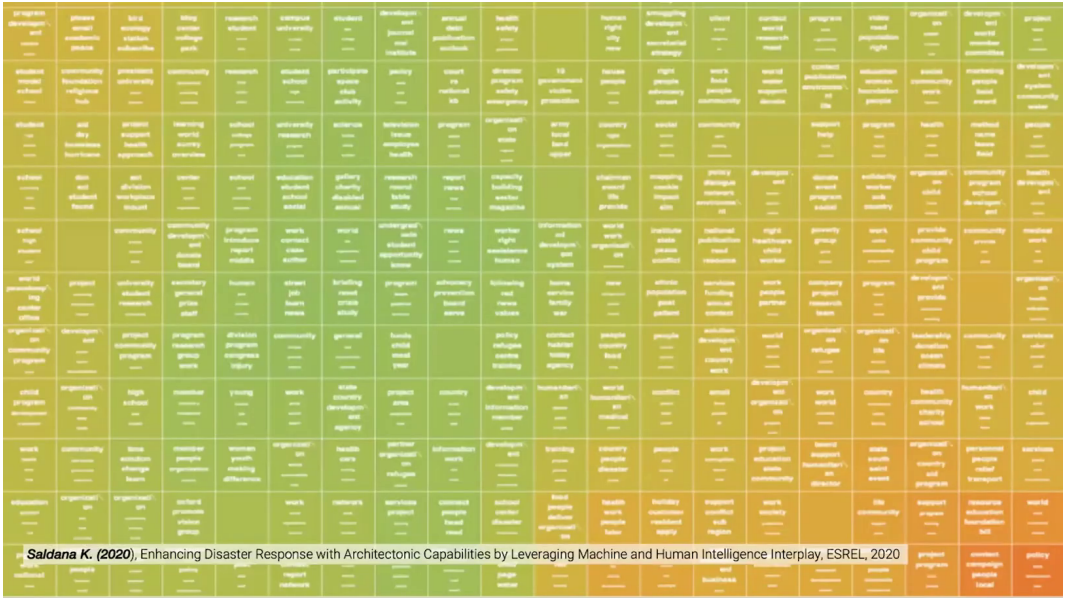

Saldana K., Bokhari, A. (2019), SOM 3D *, ETH ITA CAAD, https://themapandtheterritory.com/som-3d


Saldana K., Bokhari, A. (2019), SOM 3D *, ETH ITA CAAD, https://themapandtheterritory.com/som-3d

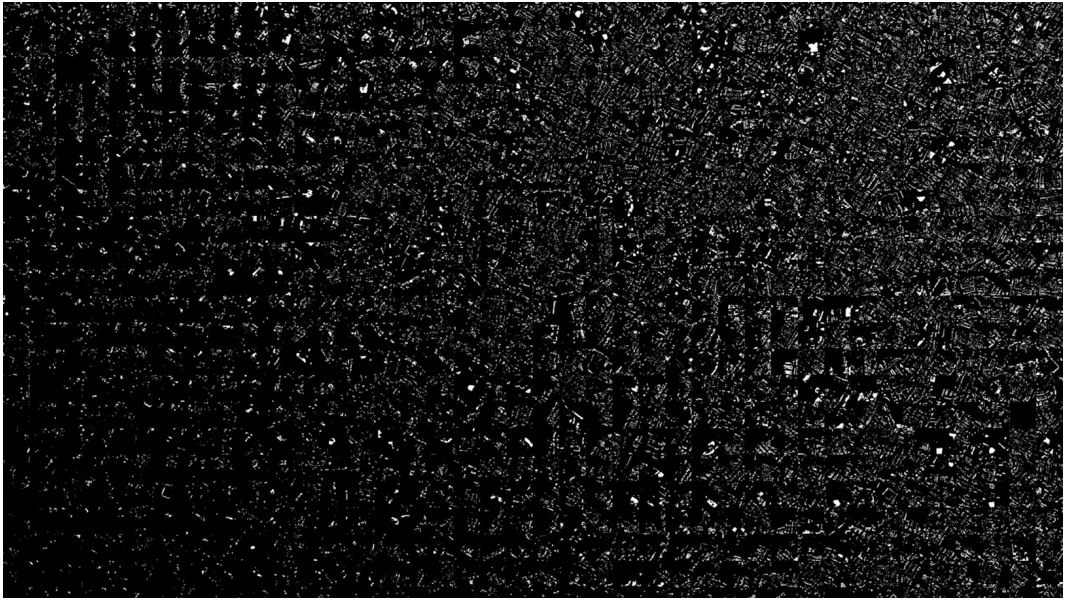
Alvarez D., Saldana K., (2017), City of indexes, Elective course 2017, ETH ITA CAAD Praxis 2018


Alvarez D., Saldana K., (2017), City of indexes, Elective course 2017, ETH ITA CAAD Praxis 2018





Saldana K. (2020), Enhancing Disaster Response with Architectonic Capabilities by Leveraging Machine and Human Intelligence Interplay, ESREL, 2020



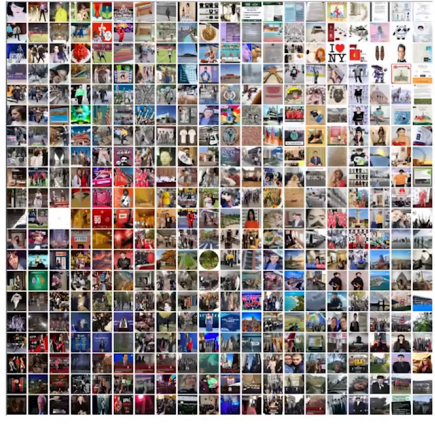
SOM ON TWEETS

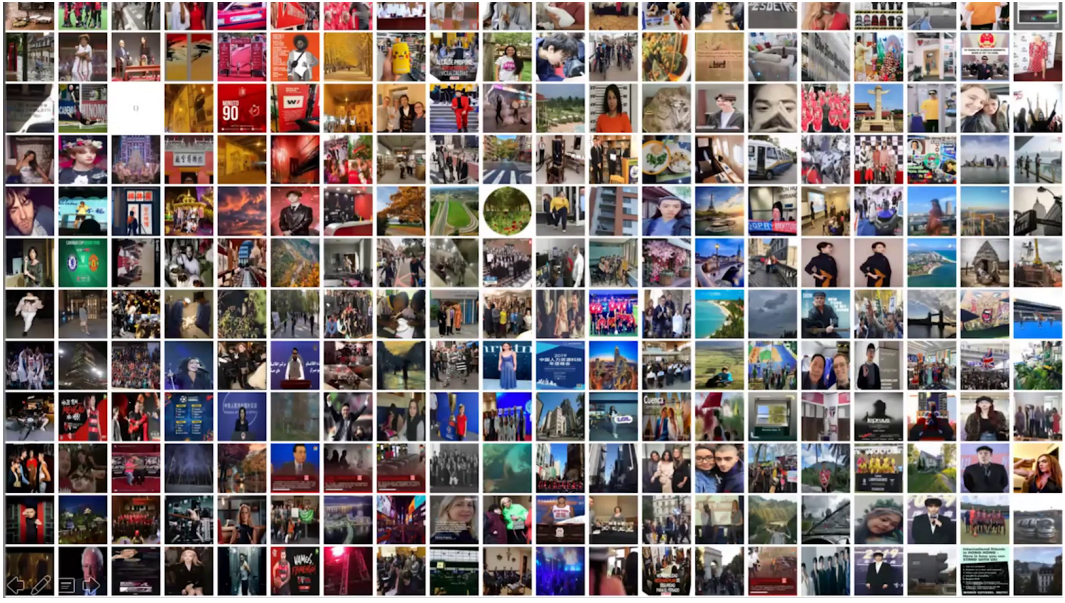
Demonstration with the previous collected tweets

image

-> "colors"

```
1600   0  1600   0  1600
   0  1600   0  1600   0
1600   0  1516  84  1536
  64  1600   0  1562  38
1477  123  1494  106 1558
  42  1600   0  1600   0
1600   0  1600   0  1512
```

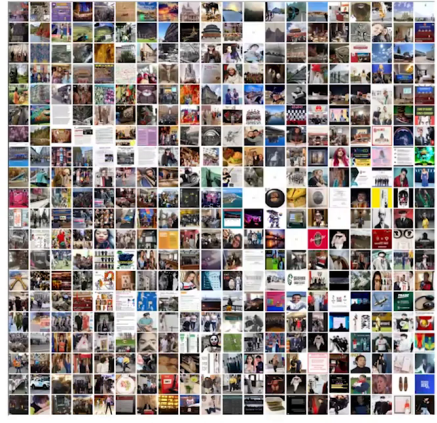


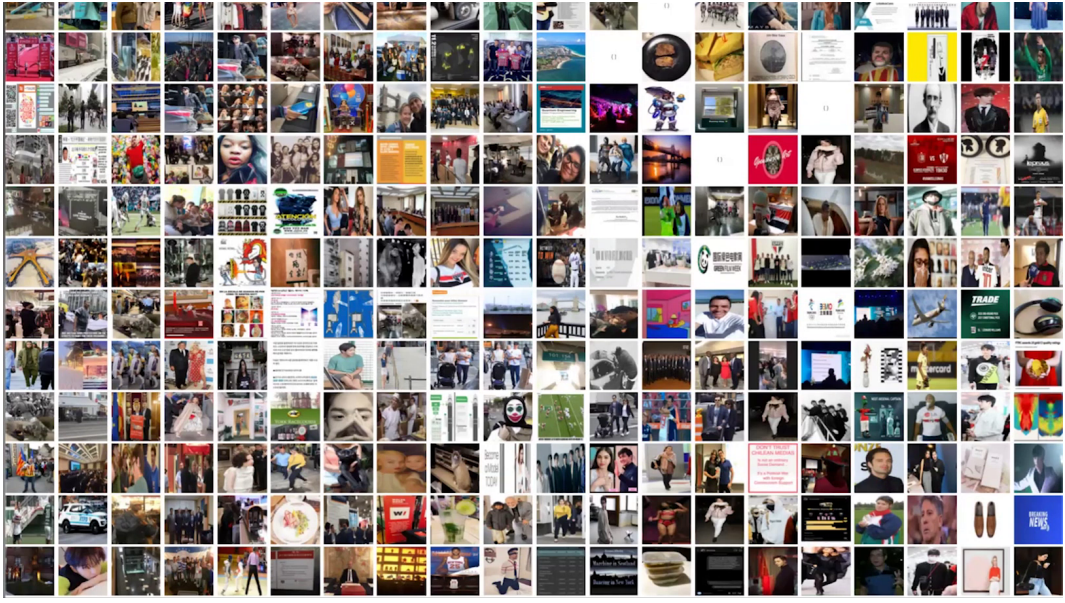


image

-> "edges"

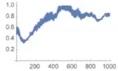
```
0.439216 0.470588 0.541176
0.694118 0.733333 0.862745
0.780392 0.811765 0.898039
0.733333 0.741176  0.74902
0.462745 0.490196 0.458824
0.505882 0.533333 0.458824
0.662745 0.694118 0.701961
```



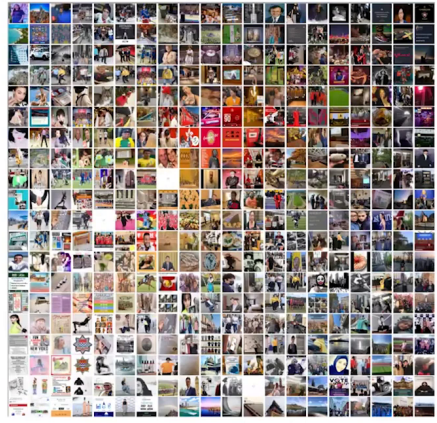


image

-> "fourier"



311.196 + 0. i	-2.844
-2.8447 - 8.53134 i	-18.802
0.0507741 - 1.67722 i	-0.8558
11.2859 + 32.356 i	2.2817
1.98793 + 0.293197 i	-2.4941
1.08693 - 0.0374585 i	-0.7286
-7.28062 + 7.17637 i	0.6167

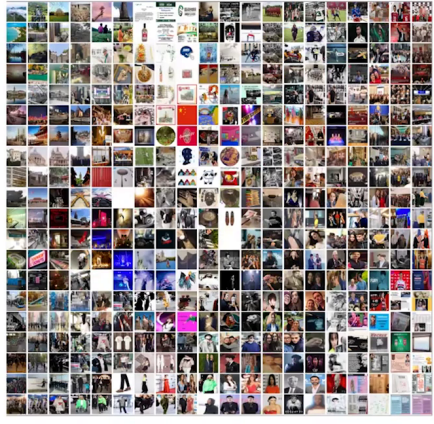


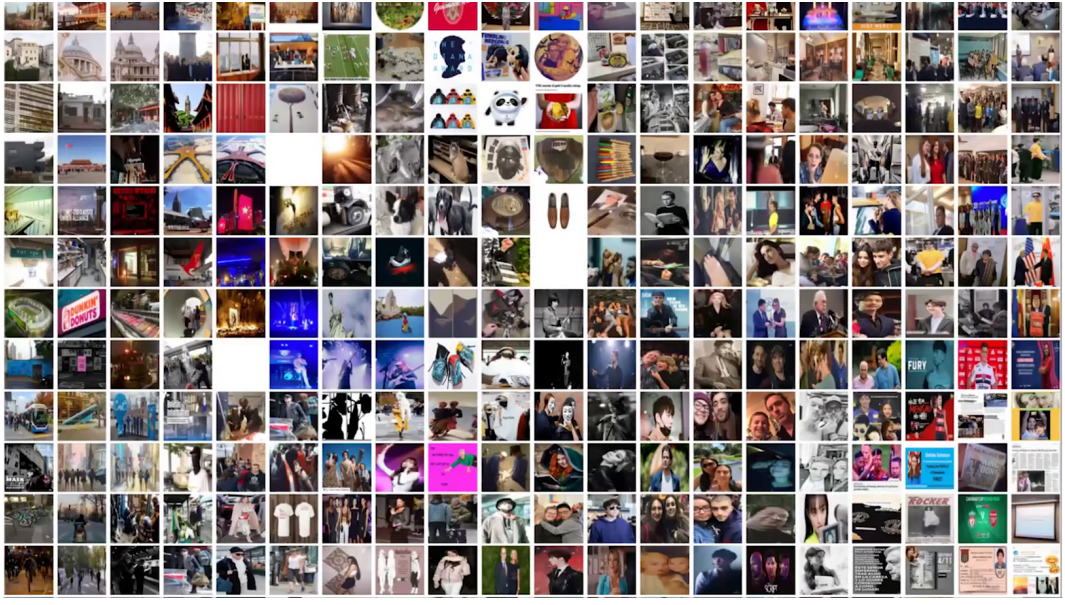
image

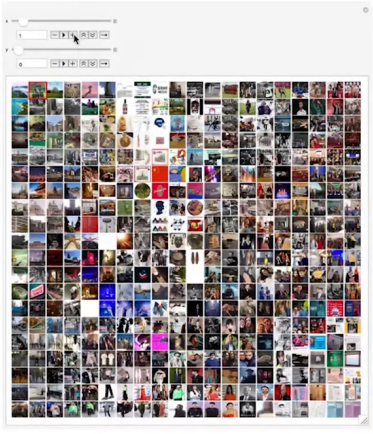
-> "feature extraction"

- common privet
- common jasmine
- laurel
- California laurel
- fruit tree
- pride-of-rochester

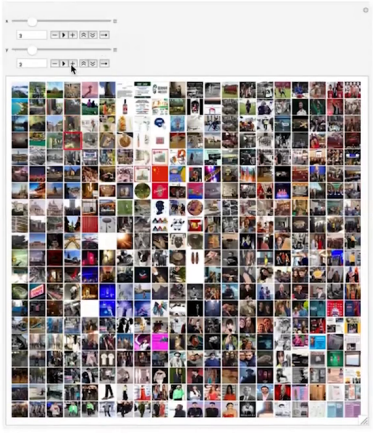
```
 8.22561  -8.62608  28.0713
-3.54004  12.2321   3.18221
-6.41089  1.24627  -2.62163
```



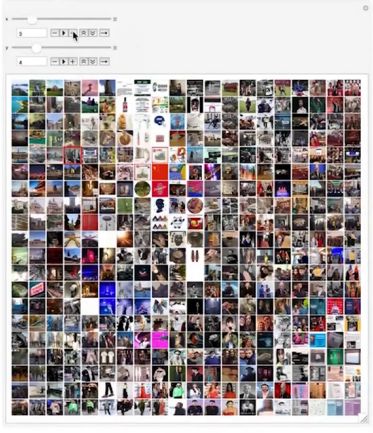

DUNKIN' DONUTS
FURY
ROCKER



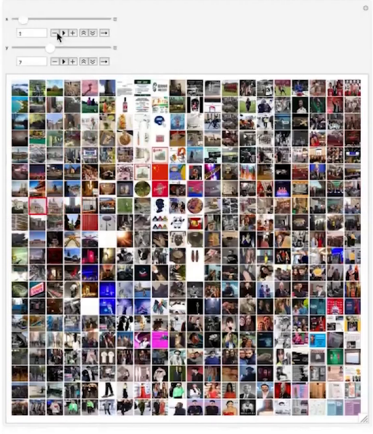
* cell N = 2 * events= 15



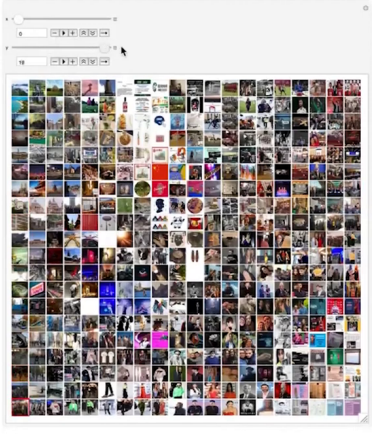
* cell N = 64 * events= 14



* cell N = 84 * events= 7



* cell N = 142 * events= 7



* cell N = 381 * events= 20

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Some resources

- A spring grid bend to fit the shape of the data.
- UbiSOM3
- Kohonen's Self-Organizing Map (SOM) Visualised
- Map&Models - Lecture 08- Self Organizing Maps